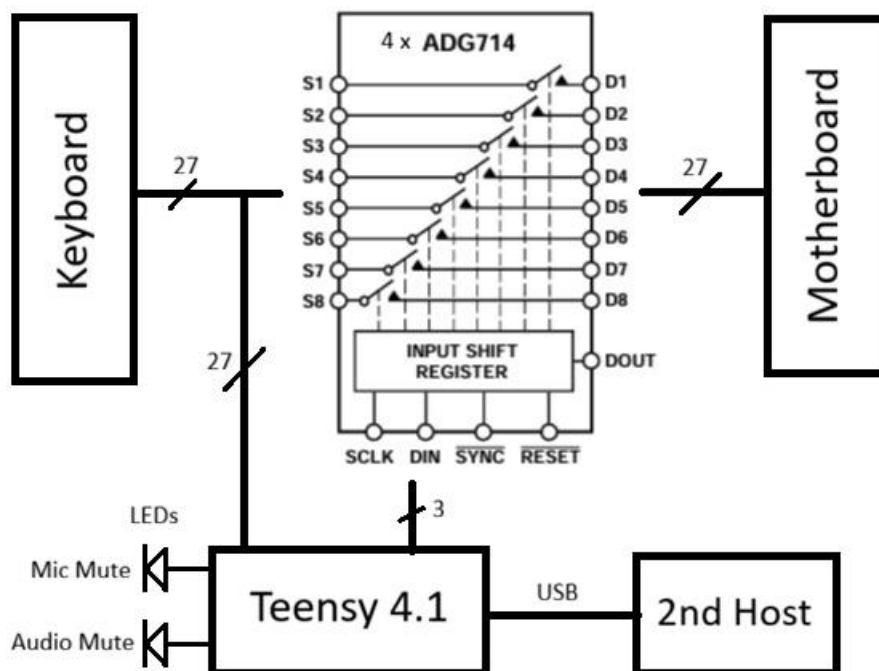


Lenovo ThinkPad P15 Gen 1 Keyboard – Dual Connection

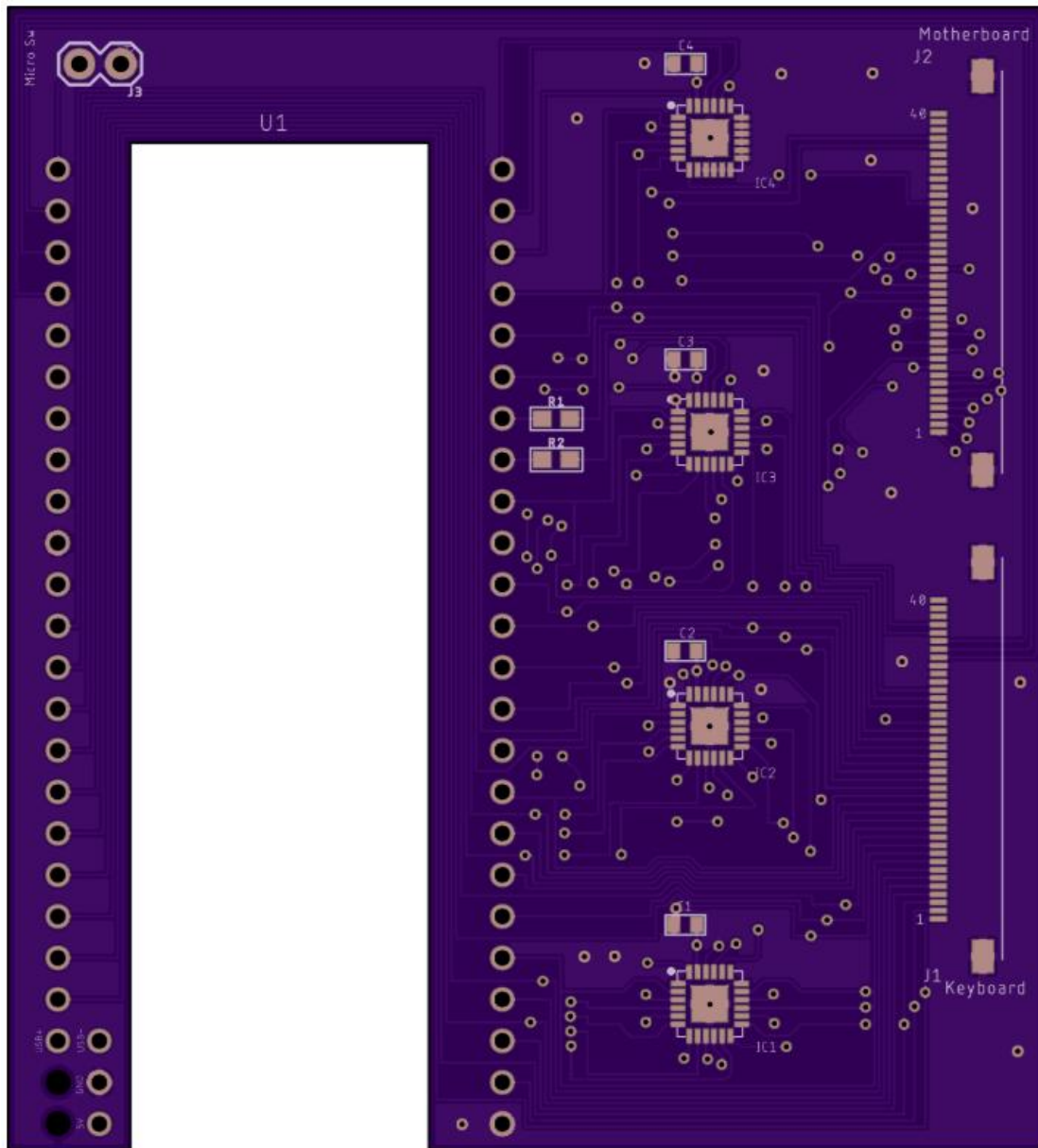
In this project, a Lenovo Thinkpad P15 Gen 1 laptop keyboard shown below is either controlled by a Teensy 4.1 that is cabled via USB to a host computer, or the row and column pins are connected to the motherboard as they would be in a normal laptop. The backlight is not used and the trackpoint is controlled by a separate circuit.



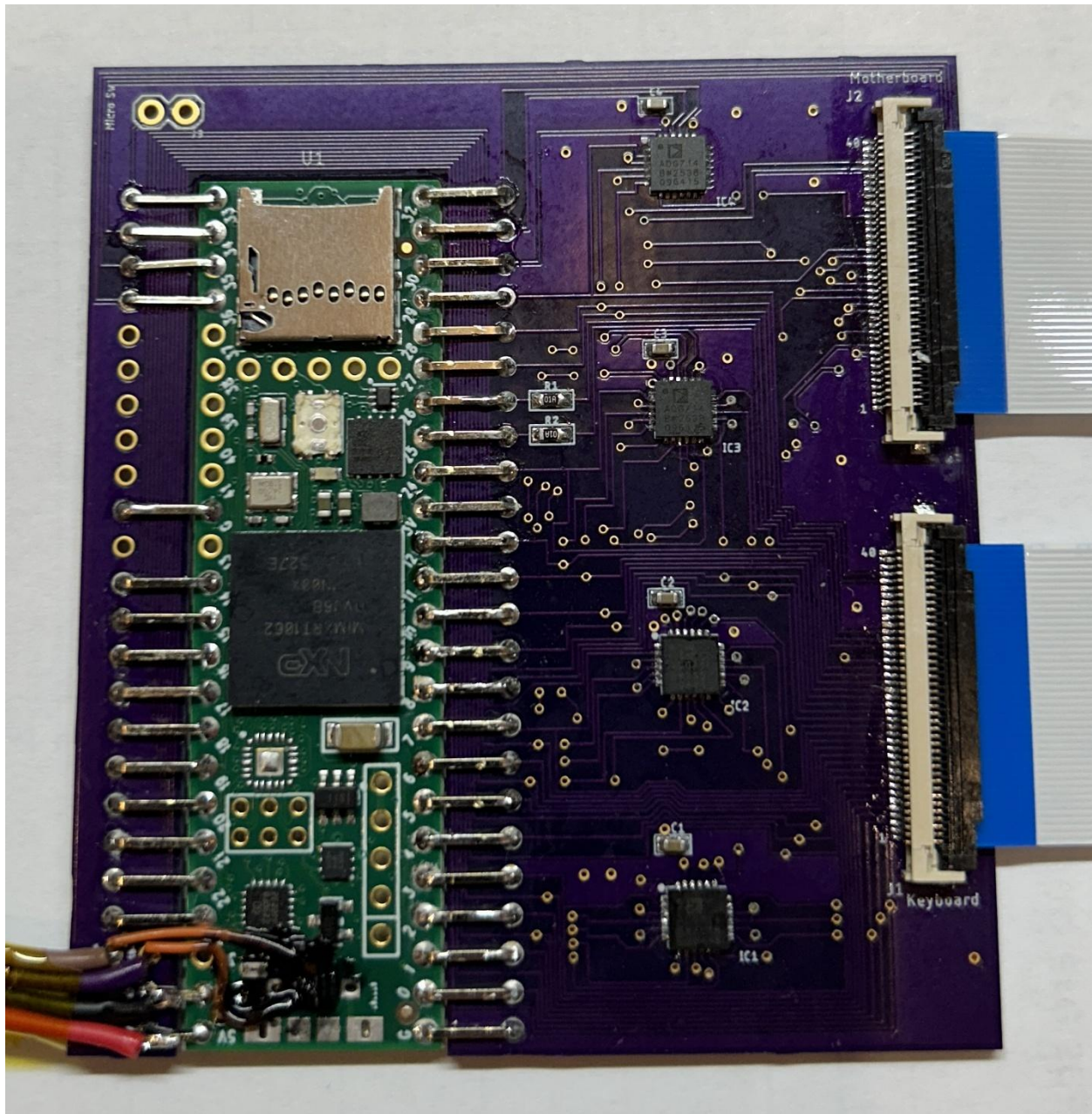
Four ADG714 octal switches connect the 27 keyboard row & column signals to the motherboard (see below). Holding the Fn key down for more than 4 seconds signals to the Teensy to switch control to the opposite host. The Teensy shifts in all 0's or all 1's to open or close the switches. The Audio Mute LED is illuminated when the Motherboard is in control and the Mic Mute LED shows the Teensy is in control.



The circuit board shown below was designed in Eagle and fabricated by OSH Park. The Teensy 4.1 sits in the cutout and is wired to the board with low profile jumpers. When the motherboard is connected to the keyboard, the Teensy configures all 27 GPIO's as inputs without pullups so they don't interfere. The Fn "Hotkey" is unique in Lenovo keyboards because it is not part of the row/column matrix. One side of the Fn key switch is always tied to ground and the other side is always available to be read by the Teensy and the motherboard. This allows the Teensy to always be able to detect when the Fn key is held down. Pads for reading an optional microswitch can be seen in the upper left corner. The micro-USB connector and push button switch have been unsoldered from the Teensy to reduce the height so the board will fit in the laptop. The 4 USB wires are soldered to pads in the lower left corner.



The assembled circuit board with all components is shown below



A hot air rework station was used to remove the micro USB connector. The rubber and plastic on the push button switch melted and was cut away with wire cutters. Equal length 30 gauge wire wrap wire was soldered to the USB D+ and D- surface mount pads (previously used by the USB connector). These wires go to the lower left of the circuit board and are coated with liquid electrical tape for protection. The USB cable is made from individual wires cut to the same length (20cm) and wrapped with Kapton tape. The USB A connector soldered to the wires does not have a plastic cover to reduce its height (see below).



The end view (see below) shows the low profile of the assembled board which allows it to fit inside the laptop case.



I already had determined the key matrix on a previous P15 keyboard project so I didn't have to run my matrix decoder routine. The 40 pin FPC pin number is listed first, followed by the connection to the Teensy GPIO numbers and the Lenovo schematic signal name.

P15/P17 Keyboard Matrix

FPC pins – GPIO # Signal Name	1-23 SENSE3	2-0 SENSE7	3-22 SENSE6	5-21 SENSE4	6-2 SENSE1	8-3 SENSE2	9-19 SENSE0	12-5 SENSE5
4-1 DRV14	L-SHIFT		R-SHIFT					
7-20 DRV0	TAB		Z	A	1	Q	TILDE	ESC
10-4 DRV4	Y	N	M	J	7	U	6	H
11-18 DRV2	F3		C	D	3	E	F2	F4
13-17 DRV1	CAPS		X	S	2	W	F1	
14-6 DRV3	T	B	V	F	4	R	5	G
15-16 DRV6	F7		PERIOD	L	9	O	F8	
16-7 DRV7	[	SLASH		;	0	P	MINUS	QUOTE
17-15 DRV5	]		COMMA	K	8	i	=	F6
18-8 DRV15			R-CNTL				L-CNTL	
19-14 DRV13		R-ALT			PNTSCN			L-ALT
20-9 DRV9	GUI	RIGHT			F12	KP .	INS	
21-10 DRV12	KP =	LEFT			END	KP [	HOME	UP
22-11 DRV10		DOWN			F11		DEL	
23-12 DRV8	BKSPC	SPACE	ENTER	\	F10		F9	F5
24-24 DRV11	KP]				PGDN	KP BK	PGUP	
36-25 DRV16	KP /	KP +	KP 9	KP 7	KP 8	KP *	KP -	NUMLK
37-26 DRV17	KP 5	KP 0	KP ENT	KP 2	KP 3	KP 6	KP 1	KP 4

The trackpoint left, middle, and right buttons are part of the keyboard and are scanned by the Teensy. The Teensy sends the button status over USB along with the keyboard key presses.

Trackpoint Middle Button = GPIO 34 on FPC pin 35

Trackpoint Right Button = GPIO 35 on FPC pin 34

Trackpoint Left Button = GPIO 36 on FPC pin 33

The keyboard, Teensy board and motherboard is shown below.

